MIL-P-0051062D(EA)
4 February 1987
SUPERSEDING
MIL-P-0051062C(EA)
16 April 1986
USED IN LIEU OF
MIL-P-51062B
18 August 1969

MILITARY SPECIFICATION

PUMP, FOG OIL, SMOKE GENERATOR, M4

This military specification has been prepared by the U.S. Army Chemical Research, Development and Engineering Center, based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-P-51062B. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers a fog oil pump for use with the smoke generator.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-12070 - Fog Oil.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1040

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STANDARDS

FEDERAL

FED-STD-123 - Marking for Shipment (Civil Agencies).

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MS24394	-	Elbow, Tube, Bulkhead, 90°, Precision Type.
MS33658	-	Fitted End, Hose Connection, Standard Dimensions For.
AN816	-	Adapter, Straight, Pipe to Tube.
AN842	-	Elbow, Pipe to Hose, 90°.
AN844	-	Elbow, Pipe to Hose, 45°.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

For M3A3 Smoke Generator:

31-15-1197	-	Pump, Fog Oil, Smoke Generator, M4, Assembly.
P31-15-1197	-	Special Packaging Instructions (SPI) for Pump, Fog Oil.

For M3A4 Smoke Generator:

31-15-1446	-	Pump, Fog Oil, Smoke Generator, M4, Assembly.
P31-15-1446	-	Special Packaging Instructions (SPI) for Pump, Fog Oil.

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 31-15-1197 or 31-15-1446, as applicable, shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the pump shall conform to the specifications and drawings listed on Drawing 31-15-1197 or 31-15-1446, as applicable.

3.2 Assembly. The pump shall be assembled as shown on Drawing 31-15-1197 or 31-15-1446, as applicable.

3.3 Leakage. The pump assembly shall not leak air when subjected to an internal pressure of 25 ± 2 pounds per square inch gage (psig) for a period of 60 ± 10 seconds when tested in accordance with 4.5.4.1.

3.4 Functioning. After a 15 minute break-in period at an air pressure of 5 + 1 psig the pump assembly shall deliver a minimum of 55 gallons per hour (gph) of fog oil when operated at a test pressure of 3 ± 1/2 psig. The pump shall operate freely and continuously (without stopping or stalling) at the test pressure for 15 ± 2 minutes, with fog oil being emitted from the ends of both the oil exhaust and oil outlet lines. No evidence of fog oil leakage shall be permitted when tested in accordance with 4.5.4.2.

3.5 Endurance. The pump shall operate continuously for 7 ± 1 hours at the test pressure specified in 3.4. The rate of delivery as specified in 3.4 shall be determined at 1-hour intervals. No evidence of fog oil leakage shall be permitted when tested in accordance with 4.5.4.3.

3.6 Preproduction. Prior to the start of regular production, a preproduction sample of pumps shall be produced for examination and test in accordance with 4.4.

3.7 Workmanship. The assembled pump shall be free from contamination (rust, grease, or foreign matter) and damage (dents, cracks, or deformation) which could affect functioning.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Precautionary. This specification involves materials which are potentially hazardous to personnel. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of these materials.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.4)
(b) Quality conformance inspection (see 4.5).

4.4 Preproduction inspection.

4.4.1 Sample. A preproduction sample of two pumps shall be produced using the same methods, materials, and equipment that will be used in regular production.

4.4.2 Inspection procedure. The pumps shall be tested in accordance with 4.5.4.1, 4.5.4.2 and 4.5.4.3.

4.4.3 Acceptance/rejection criteria. The preproduction pumps shall meet all the tests of this specification to be acceptable. Pumps that do not operate freely and continuously are not acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.5 Quality conformance inspection.

4.5.1 Lotting. A lot shall consist of the pumps of one interfix number, produced by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions.

4.5.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. For endurance test general inspection level 1 shall be used.

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4.5.3 Inspection procedure.

4.5.3.1 For examination and nondestructive test. Sample components, pump assemblies and packaging shall be examined in accordance with the classification of defects and with MIL-STD-105. For endurance test use an AQL of 2.5 percent defective.

4.5.3.2 Critical defects. Each pump in the lot shall be examined for the critical characteristics listed in the classification of defects.

4.5.3.3 Classification of defects.

(a) Pump, fog oil, smoke generator, M4 (Drawing 31-15-1197 or 31-15-1446, as applicable).

Categories	Defects	Acceptance standards
Critical:		
1	Leakage	4.5.4.1
2	Functioning	4.5.4.2
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Identification plate missing or incorrect	
103	Finish missing or incorrect	
104	Fog oil outlet thread incorrect (AN816-10J)	
105	Outside configuration of oil exhaust incorrect (AN842-10D/MS 33658)	
106	Pressurizing air inlet thread (3/4-16 UNJF-3A) incorrect (MS 24394-8)	
107	Outside configuration of oil inlet connector incorrect (AN844-12D/MS33658)	
*108	Clearance 0.025 (position A&B) incorrect	
109	Bolt torque incorrect	
110	Workmanship (3.7)	

*Check prior to final assembly.

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(b) Packaging inspection (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Surfaces not clean
102	Openings not covered as specified
103	Internal fiberboard blocking insufficient to make a tight pack
104	Container damaged (broken, torn or punctured)
105	Marking incorrect, missing, or illegible
106	Workmanship (3.7)

4.5.4 Tests. Tests shall be conducted as follows:

4.5.4.1 Leakage. Plug the air intake and oil outlet ports of the pump assembly. Connect the oil inlet and oil exhaust ports to a regulated source of dry filtered compressed air with in line calibrated pressure gauge and shut-off valve. Subject the pump assembly to the required pressure. Close shut-off valve. Observe the pressure gauge for the required period of time for any drop in pressure. No drop in pressure is allowed. A drop in pressure indicates leakage. Do Not Submerge In Water.

4.5.4.2 Functioning. Mount the pump assembly on a suitable stand and connect 7 to 10-foot lengths of hose or tubing to the oil inlet, oil outlet, and oil exhaust ports. The oil inlet hose shall contain a strainer conforming to Drawing 31-15-1188. The oil outlet hose shall contain a metering orifice with a 0.180 ± 0.010-inch diameter by 1/8 + 1/32 inch long hole. Place the hose or tubing lines into a container of fog oil, type SGF2, MIL-F-12070. Place the end of the inlet line below the surface of the fog oil and the ends of the outlet and exhaust lines above the surface of the fog oil. Connect a regulated source of dry filtered compressed air to the pump assembly air inlet port. Open the pump assembly bleed valve and introduce air into the pump at the required break-in pressure. When air exhausts from the bleed valve, close the valve and operate the pump for a minimum break-in period of 15 minutes. Without stopping the pump, reduce the pressure to the required test pressure. Place the outlet hose into a calibrated container to determine the volume of oil delivered by the pump in gallons per hour (gph). Record the rate of delivery and operating pressure at end of test. During test observe for any fog oil leakage. No leakage is permitted.

4.5.4.3 Endurance. Operate the pump in accordance with 4.5.4.2 for the required time. Record the rate of delivery and operating pressure at the required intervals. During test observe for any fog oil leakage. No leakage is permitted.

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5. PACKAGING

5.1 Interplant shipment (see 6.2). The pump shall be unit packed and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in accordance with FED-STD-123, and shipping containers shall include the contractor's name, address, contract number, and date of manufacture.

5.2 Repair parts. When this item is being procured for storage and issue as a repair part, unit packing, packing and marking shall be as specified in the applicable SPI, P31-15-1197 or P31-15-1446.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Whether interplant shipment or repair parts packaging is required.
(c) Preproduction:

(1) Time allowed for contractors to submit samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether to proceed with production.

6.2 Interplant shipment. Packaging and packing for interplant shipment is for supplies and materials that do not enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

6.3 Subject term (key word) listing.

Fog oil
Pump, fog oil
Smoke generator

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1040-A182

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-0051062D(EA)	PUMP, FOG OIL, SMOKE GENERATOR, M4

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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